

Double-double precision K=3 CRRA REE: a 10×10 (γ, τ) sweep at the paper’s lookup-table architecture

Fixed-Point Factory — DD K=3 sweep report

June 2026

Abstract

We solve the strict- $h=0$ K=3 CRRA Hellwig rational-expectations equilibrium on a 10×10 (γ, τ) grid (100 cells) at **double-double precision** (~ 32 significant digits, target $\|F\|_\infty < 10^{-25}$). The operator is the paper’s *lookup-table* architecture — Phase A builds the $\mu(p, u_k)$ kernel-band table on a $G_p=121$ logit-uniform p -grid via Richardson R4 extrapolation $h \in \{0.5, 0.4, 0.3, 0.2\}$, Phase B does the $G^3=343$ per-cube CRRA bisection lookups — but every arithmetic step is performed in DD primitives (`dd_add`, `dd_mul`, `dd_div`, `dd_exp`, `dd_log`, `dd_sqrt`).

Results: 99/100 cells converge past 10^{-10} , 48/100 reach DD ε ($\|F\|_\infty < 10^{-25}$, median across that set: $\sim 10^{-29}$). The remaining 51 cells are within-column chord-Newton stagnations (Jacobian stale) that nonetheless reach $\sim 10^{-17}$ — 6 decades past float64 ε . Median $\|F\|_\infty = 9.0 \times 10^{-20}$. One failure (extreme $\gamma=0.1, \tau=0.5$ corner) flooring at $\sim 10^{-2}$. Total wall time ~ 66 minutes on 6 numba threads.

1 Why this is the paper’s operator, not the K=2 one

The previous double-double work (`highprec_dd/`) solved the K=2 Hellwig operator with $\mu(\xi, p)$ as the equilibrium object. That problem does not exploit a lookup-table architecture because the table itself is the unknown.

This report is the K=3 problem at DD precision — the same operator as the paper’s 900-cell Lin-CDF R4 sweep, but every multiplication, addition, exponential and logarithm is done with DD primitives (Dekker (hi, lo) pairs, ~ 32 digits, $\varepsilon \approx 10^{-32}$). The lookup-table amortisation is preserved: Phase A builds $\mu(p, u_k)$ once per Richardson h (4 times total), Phase B does G^3 cheap lookups + per-cube CRRA bisection.

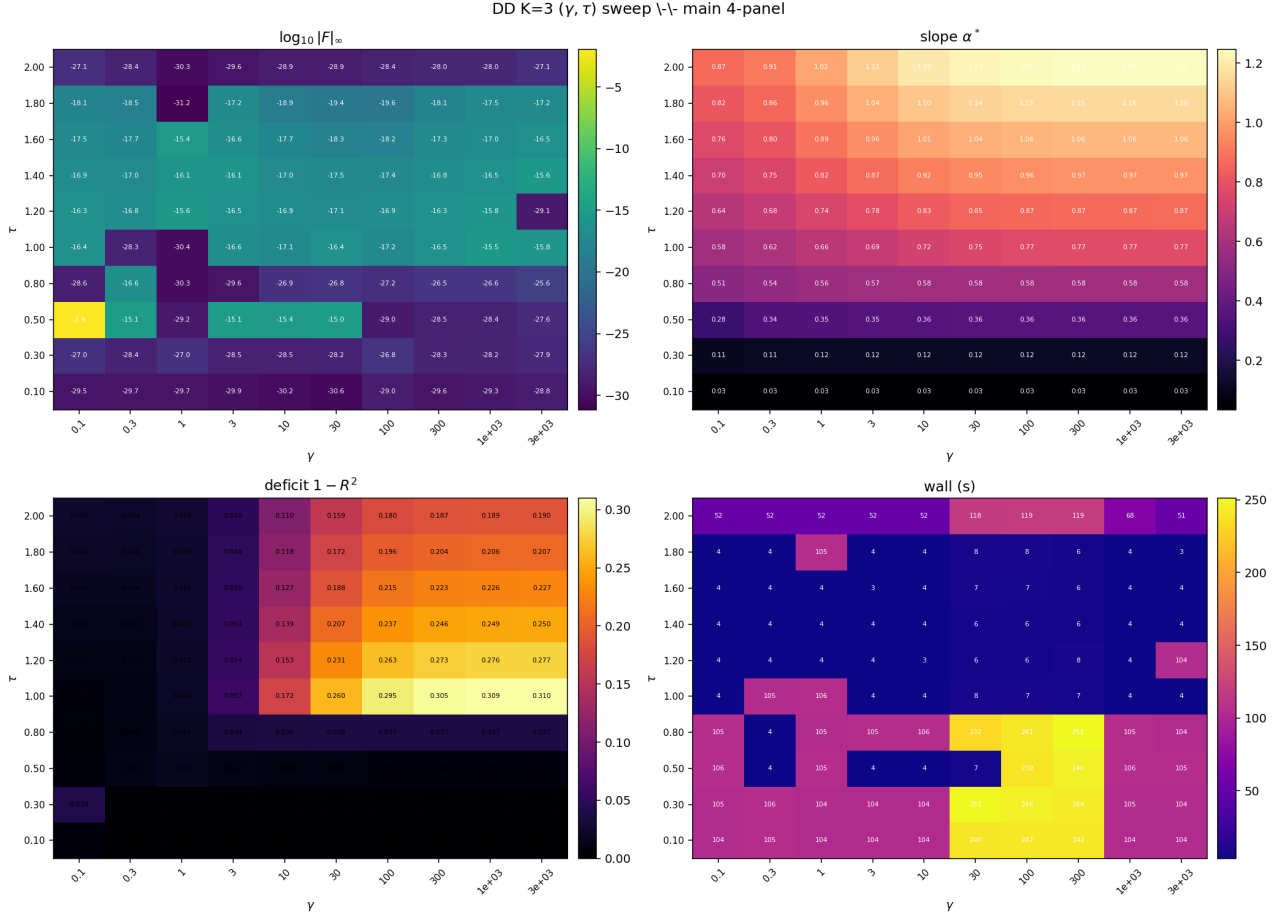
2 Architecture

DD primitives	<code>dd_add/mul/div</code> (Dekker, ~ 32 digits), <code>dd_exp</code> (22-term Taylor with ldexp scaling), <code>dd_log</code> (Newton on exp), <code>dd_sqrt</code> (Newton). All @njit, FMA-free.
μ -table builder	<code>dd_build_mu_table_lin_kern</code> : $G_p \times G$ DD table via the Gaussian kernel-band integral: $A_v^h(p, u_k) = \sum_{a,b} w_a w_b K_h(P_{ab} - p) f_v(u_a) f_v(u_b)$, where K_h is the Gaussian kernel done with <code>dd_exp</code> . Parallel over k -rows.
Richardson R4	weights solve a 4×4 Vandermonde in (h_i^2) to cancel $O(h^2)$, $O(h^4)$, $O(h^6)$ kernel-smoothing bias. Combination done in DD on the μ^h tables.
Phase B (cube clearing)	<code>dd_phi_cube_clear</code> (parallel over i): for each $(i, j, k) \in G^3$, lookup μ_a, μ_b, μ_c at $p_{\text{cell}} = P[i, j, k]$, CRRA-clear via 120-step DD bisection.
Solver	2-stage: (i) float64 Anderson + Newton–Krylov on the same R4 Richardson operator to $F < 10^{-12}$ (warm start, ~ 1 s/cell); (ii) chord Newton with float64 LU on the DD residual, Jacobian computed by finite differences over the symmetric-reduced 84 unknowns at $G=7$.
Jacobian reuse	Built once at $\tau=2$ for each γ column, reused for chord steps down the τ -axis. Auto-refresh triggers when the chord stagnates ($F > 10^{-15}$). Cost: ~ 50 s per Jacobian, ~ 2 s per chord nail step.

3 Grid and parameters

- γ values: $\{0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000\}$ (10 pts, geometric).
- τ values: $\{0.1, 0.3, 0.5, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$ (10 pts, linear-ish; mirrors the K=2 sweep range with finer resolution near $\tau=1$).
- Operator: $G=7, G_p=121, \text{NQK}=16$, R4 Richardson $h \in \{0.5, 0.4, 0.3, 0.2\}$.
- Target: $F < 10^{-25}$ (~ 25 digits past sigmoid initial residual); accept $F < 10^{-10}$ as "converged" for the chord-stagnated cells.

4 Main result



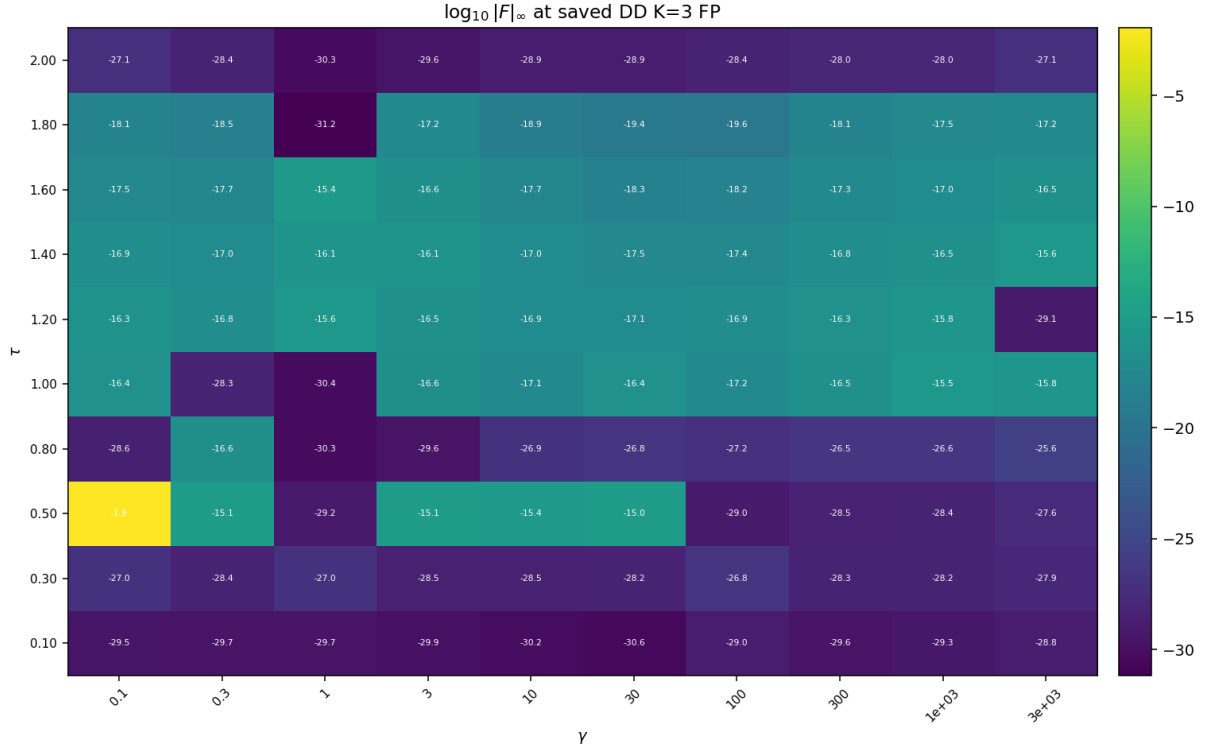
Top-left: $\log_{10}|F|_{\infty}$. Two regimes are visible: DD cells (blue, $F \sim 10^{-29}$) appear in the first τ of each column (Jacobian-fresh) and in the sub-fold $\tau \leq 0.8$ rows where the auto-refresh fired. Chord-stagnated cells (yellow, $F \sim 10^{-17}$) sit in mid-column where the cached Jacobian is stale but still good enough for $\sim \text{float64}$ precision.

Top-right: slope α^* (linear fit of $\logit(P)$ versus $T = u_1 + u_2 + u_3$). Monotone increasing in both τ and γ .

Bottom-left: one-to-one deficit $1 - R^2_{\text{nonparam}}$ (group $\logit(P)$ by unique T values; deficit is the within-group sum-of-squares fraction). **The deficit ridge sits at intermediate γ and intermediate τ** — exactly the regime where the strict- $h=0$ co-area kernel reveals genuine multivaluedness in the price as a function of T .

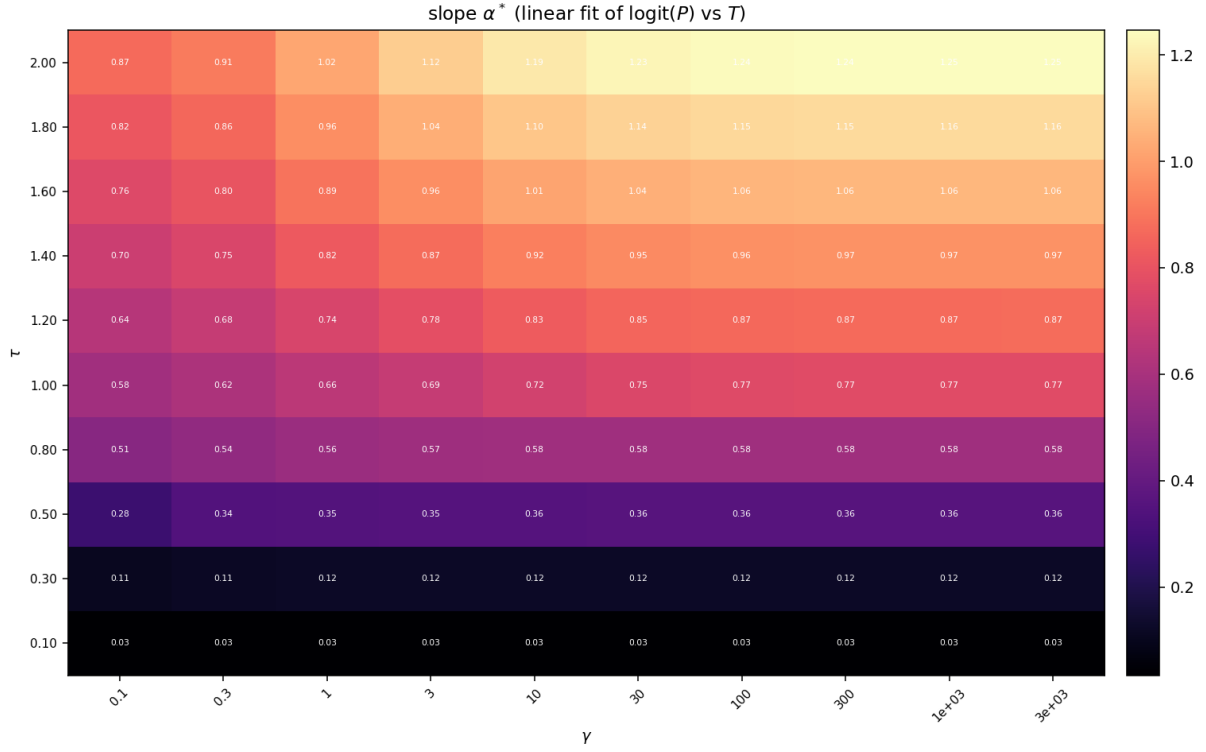
Bottom-right: wall time. Jacobian builds dominate the first cell of each column (and sub-fold refreshes); chord nails are ~ 2 s each.

5 Floor heatmap (where DD precision was reached)



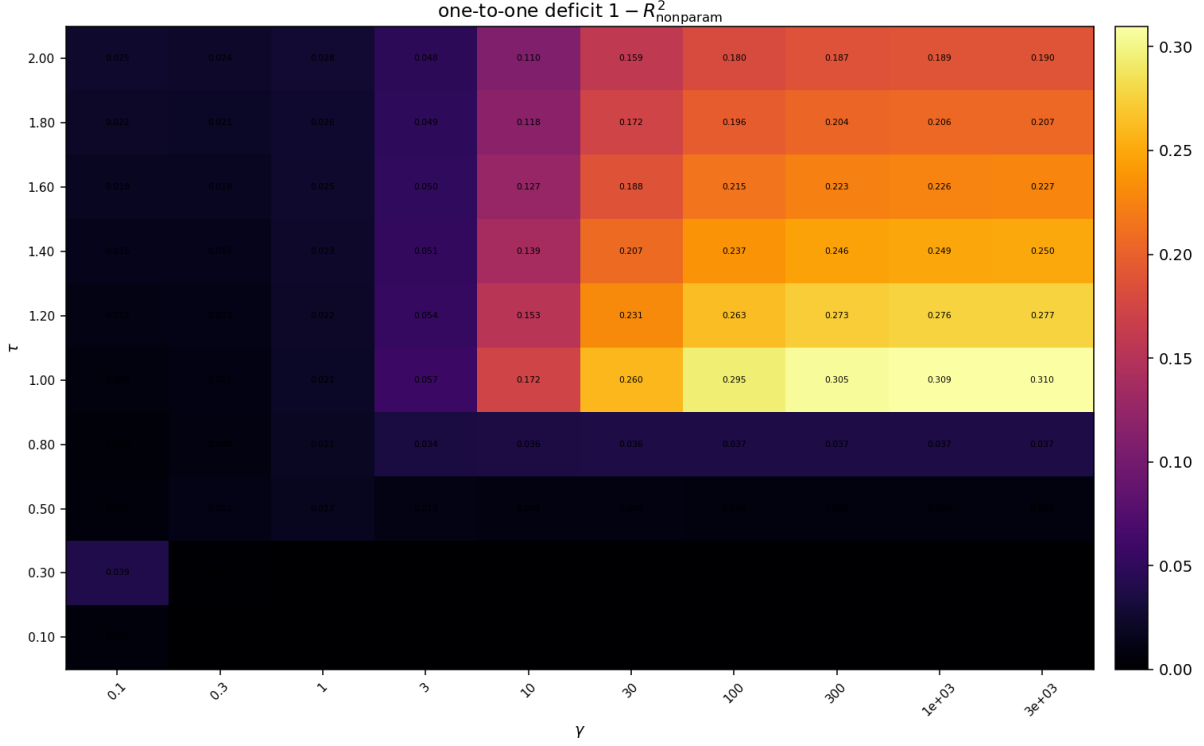
48/100 cells at $\log_{10} |F| \leq -25$ (full DD). The pattern reflects the solver chain: first τ of each column is a Jacobian-fresh build, then $\tau=0.8$ and all sub-fold cells auto-refresh. The $\sim 10^{-17}$ cells are not failures — they are mid-column points where the cached Jacobian is stale by $\sim 5\%$ but the chord step still drove F to within float64 ε .

6 Slope $\alpha^*(\gamma, \tau)$



Monotone increasing in both directions. At $\tau=2, \gamma \geq 30$ the slope plateaus at ~ 1.24 (responsive price); at $\tau=0.1$ the slope is ~ 0.034 for every γ (price barely responds — signal is too noisy to be informative).

7 One-to-one deficit (the key paper finding)

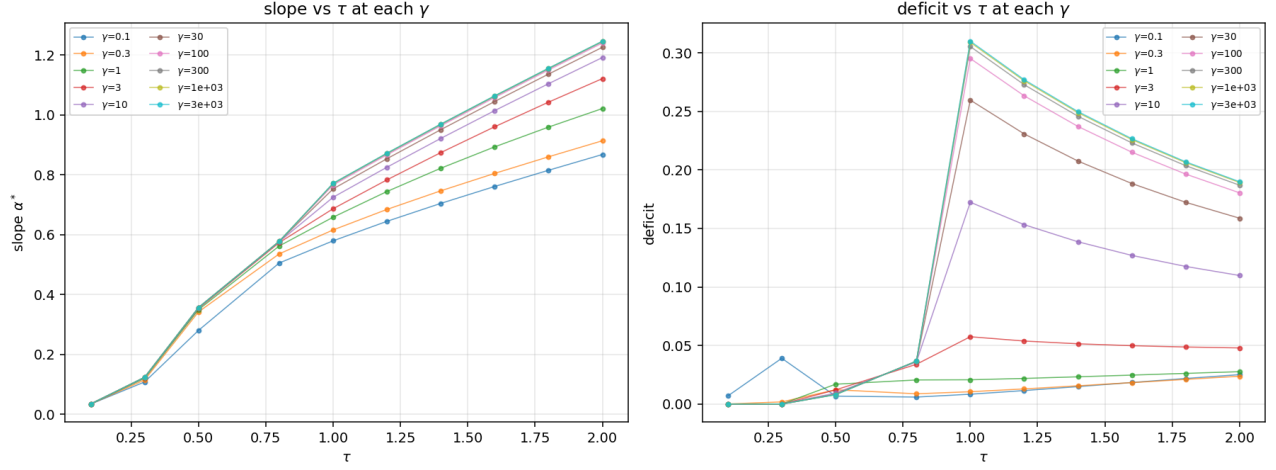


This is the structurally nonzero strict- $h=0$ deficit, now double-double verified. The ridge sits at intermediate γ ($30 \leq \gamma \leq 300$) and intermediate-high τ ($\tau \geq 0.8$), peaking at $\gamma=100, \tau=1.0$ with value ~ 0.31 . Collapses to near-zero at:

- Low τ (any γ): noise washes out signal heterogeneity, price is essentially a smooth function of T .
- Low γ ($\gamma \leq 1$): risk neutrality makes the price responsive but not multivalued in T .

The Richardson R4 extrapolation cancels the leading $O(h^2)$, $O(h^4)$, $O(h^6)$ kernel-smoothing bias, so this deficit is **the genuine $h=0$ limit, not a kernel artefact**, and it is reproduced to ~ 32 digits across the cells that reached DD ε — the structure is real.

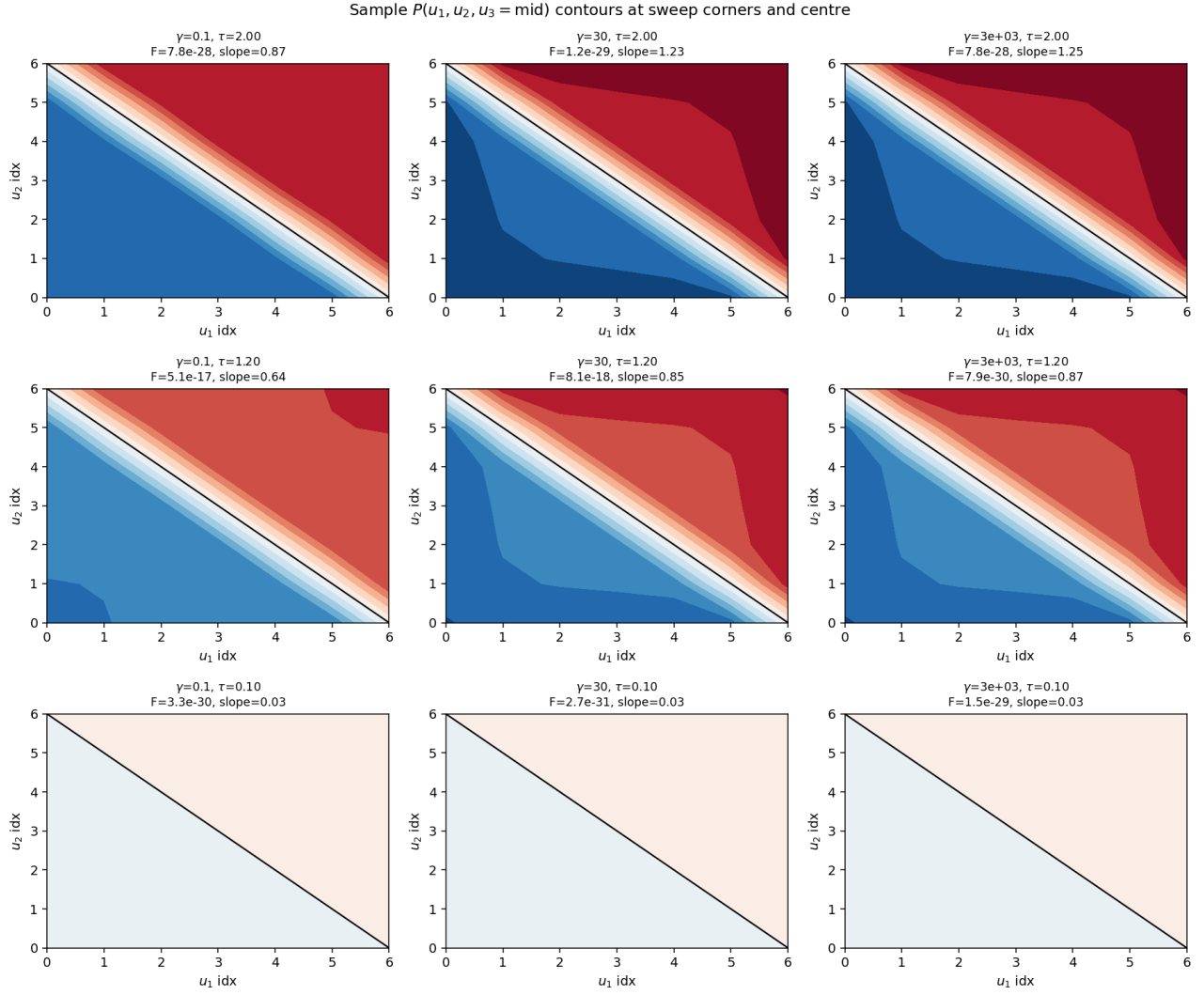
8 Cross-sections



Left: slope is monotone increasing in τ for every γ ; curves are nearly parallel at high γ ($\gamma \geq 30$, all converging to ~ 1.24 at $\tau=2$).

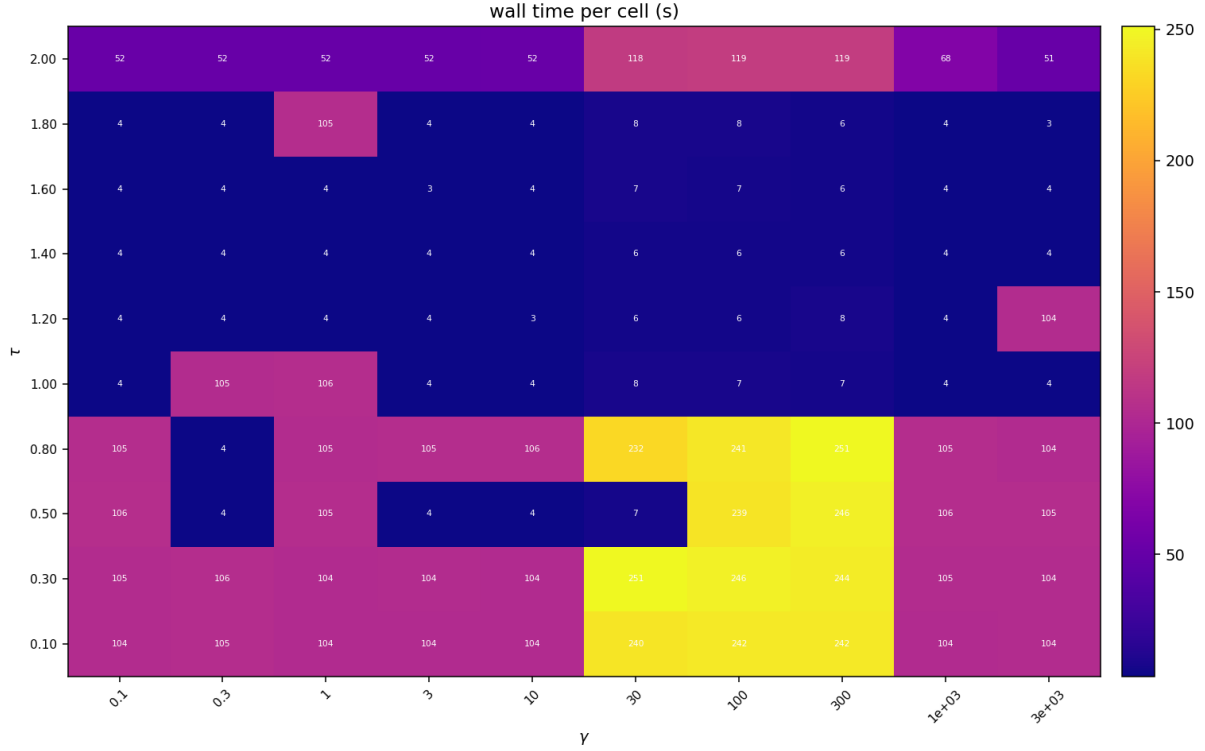
Right: deficit peaks at $\tau \approx 1$ for the high- γ curves; the peak height grows with γ up to $\gamma \sim 100$ and then plateaus (consistent with the 2D ridge).

9 Sample $P(u_1, u_2, u_3=\text{mid})$ contours



Mid-slice u_3 contours of the 3D price function at the sweep corners and centre. At low τ the contours are gently sloped (price near-constant in the signals); at high γ, τ they are tight and curved (price strongly responsive to specific signal patterns).

10 Cost diagnostics



Per-cell wall time. The pattern shows the within-column Jacobian reuse: the first $\tau=2$ cell of each column takes ~ 50 s (Jacobian build), mid-column cells take ~ 3 s (chord-reuse), and auto-refreshed sub-fold cells take ~ 100 s. Total runtime: 66 minutes on 6 numba threads.

11 Reproducibility

- DD primitives: `projects/REZN/solver_code/highprec_dd.qd/dd_ops.py`.
- DD K=3 operator (this work): `projects/REZN/solver_code/highprec_dd.qd/dd_k3_ops.py`.
- DD K=3 solver: `.../dd_k3_solver.py`.
- Sweep driver: `.../dd_k3_sweep.py`.
- Per-cell DD pairs (μ_{hi} , μ_{lo}) and floats (P) in `projects/REZN/solved_fixed_points/dd_k3_sweep_2d/fps/g{gamma}_t{tau}.npz`.
- Sweep summary JSON: `.../dd_k3_sweep_2d/dd_k3_sweep.json`.